Lekkala Ganesh

AI/ML Engineer | LLM Applications, RAG & Data Science
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Summary

AI/ML engineer building **LLM applications** and **data-science models** in **Python**: **RAG** pipelines (**chunking**, **embeddings**, **vector search**), **conversational AI** — **chatbots** and **voicebots** with **speech-to-text** and **text-to-speech** — **LangGraph agents** with guardrails, and the **REST APIs** behind them, shipped with **statistical evaluation** and **structured logging**. **Automation Testing** (500+ automated tests) and **CI** across public repos. B.Tech CSE.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML5, CSS3
Machine Learning & Statistics: scikit-learn, Pandas, NumPy, SciPy, Regression, Classification, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation, EDA, TF-IDF, Cohen's Kappa, Wilcoxon Signed-Rank, Class Imbalance, Precision/Recall
Generative AI & Agents: GPT-4o, Gemini, Ollama (local inference), Hugging Face Transformers, PyTorch, LangChain, LangGraph, MCP, Tool Calling, Guardrails, Prompt Engineering, Multi-Provider Fallback, SSE Streaming
RAG & Conversational AI: Embeddings (sentence-transformers), Chunking, Vector Search (LanceDB, pgvector, ChromaDB), Semantic Search, Chatbots, Voicebots, Speech-to-Text (Whisper, Web Speech API), Text-to-Speech, Intent Classification
Backend & Data: FastAPI, Flask, REST APIs, OpenAPI/Swagger, JSON Schema Validation (Pydantic, Zod), WebSockets, MERN (MongoDB, Express.js, React, Node.js), PostgreSQL (Row-Level Security, Triggers, Indexes), SQLite, Supabase, Schema Design, Joins & Aggregations
Engineering & Ops: Automation Testing, pytest, Vitest, Playwright, GitHub Actions (CI), Docker, Git, Linux, Structured Logging, Correlation IDs, Token-Cost Tracking, ONNX Runtime, Vercel

Experience

Freelance AI/ML & Full-Stack Engineer — Remote
Self-Employed — *Aug – Nov 2025, Mar 2026 – Present*

- **Client Websites:** Built **2 responsive, accessible** sites (**HTML5/CSS3**, mobile-first, **Core Web Vitals**, **SEO**) for U.S. construction clients — Ultracore Construction, McAllister Builders — running sign-off calls across time zones.
- **Marketplace Platform (Ongoing):** Building Aglaowner, a **Next.js** self-serve small-business marketplace — mobile **OTP** and **DigiLocker** identity verification, **WhatsApp**-based contact unlock, listing search/filter across 800+ live listings.
- **Fintech (Completed):** Built IQCommune, a **Next.js** site with budgeting, retirement, and portfolio calculators.

Junior Software Developer — Mumbai, India (Remote)
AlterSquare (OrderStack Partnership) — *Dec 2025 – Mar 2026*

- Delivered **9 client web apps** to production in **90 days** with no critical bugs, incl. the 2A Construction site for a U.S. client.
- Built **LLM fallback** (GPT-4o → Gemini → OpenRouter) for VoxAI **SSE** voice streaming and SunScope Pro 3D shadows.
- Implemented **Supabase/PostgreSQL** backends with **Row-Level Security** and **JWT** auth for a finance tracker and a contractor platform; built to **WCAG 2.1 AA** with custom a11y hooks and automated **Playwright** tests.

Web Developer — Ahmedabad, India
Dash Technologies Inc. — *Jan 2025 – Jul 2025*

- Built **12+ reusable React components** for a **HIPAA-aligned** patient portal (appointments, patient history), cutting accessibility audit findings by **40%**; verified on Chrome, Firefox, Safari, and Edge.

Full Stack Web Development Intern — Remote, Part-time
Slash Mark IT Solutions — *Dec 2024 – Apr 2025*

- Delivered **6 full-stack projects** on the **MERN stack** (MongoDB, Express.js, React, Node) — including a URL Shortener **REST API** and a Food Delivery app with **authentication**, geolocation and payment flow.

Machine Learning Intern — Virtual
Bharat Intern — *Oct 2023 – Nov 2023*

- Built **regression** and **classification** models in **Python** with **scikit-learn** and **Pandas** — **data cleaning**, **EDA**, **feature engineering**; compared estimators on held-out metrics via **cross-validation** and **hyperparameter tuning**.

Projects

Cost-Efficient RAG Service | *Python, FastAPI, LanceDB, sentence-transformers, litellm* — GitHub

- **Chunking** → **embeddings** (all-MiniLM-L6-v2) → **LanceDB vector search** → generation, with a second **pgvector** backend; **0.71 Recall@5**, **34.6 ms p95**, documented **token-cost/TCO** comparison.

LLM-as-Judge Evaluation Pipeline | *Python, scikit-learn, SciPy, Rubric Design* — GitHub

- Measures 5 judge biases (position, verbosity, self-enhancement, sycophancy, score clustering) using **Cohen's kappa** and **Wilcoxon signed-rank** tests against **105 hand-labeled** human judgments; **200 tests**.

ATS Resume Optimizer | *Python, scikit-learn, Streamlit, GitHub Actions* — GitHub

- Resume-to-job **semantic matching** via **scikit-learn TF-IDF** and **cosine similarity** with keyword-gap extraction; **276 pytest tests** and **GitHub Actions CI** (3.11/3.12, ruff) over **∼20K lines of Python**.

Education

Parul University — Vadodara, Gujarat
Bachelor of Technology in Computer Science & Engineering; GPA: 7.53/10 — *Jul 2022 – Jun 2025*

Guntur Engineering College — Guntur, Andhra Pradesh
Diploma in Mechanical Engineering; Marks: 84.12% — *2019 – 2022*